



Compressible convection in cartesian coordinates with Underworld3

Juan Carlos Graciosa¹ 

¹Australian National University

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1. INTRODUCTION AND BENCHMARKING

The convection of the Earth's mantle is usually modelled as an incompressible process, referred to as the Boussinesq approximation. However, in the Earth's mantle, the pressure increase associated with depth also increases the density due to self-compression (King et al. 2010). In some applications, this compressibility may be non-negligible and modelling it may be desirable. Over the years, several mantle convection formulations have been devised to take this compressibility into account. One such formulation is the Truncated Anelastic Liquid Approximation (TALA) (Ita and King, 1994; Jarvis and McKenzie, 1980). In this approximation, a depth-dependent reference profile is defined which dictates how the density varies with depth. This is subsequently used to calculate the volumetric changes as a result of pressurization and heating (Gassmüller et al., 2020). In addition, the buoyancy term in the momentum equation precludes the contributions of the density changes due to pressure variations (van Zelst et al., 2022). Taking compressibility into account adds temperature gradients and adiabatic density which increases the complexity of mantle dynamics (Tan and Gurnis, 2007). Here, we easily implement an isoviscous compressible convection model with the Truncated Anelastic Liquid Approximation (TALA) formulation in cartesian coordinates using Underworld3.

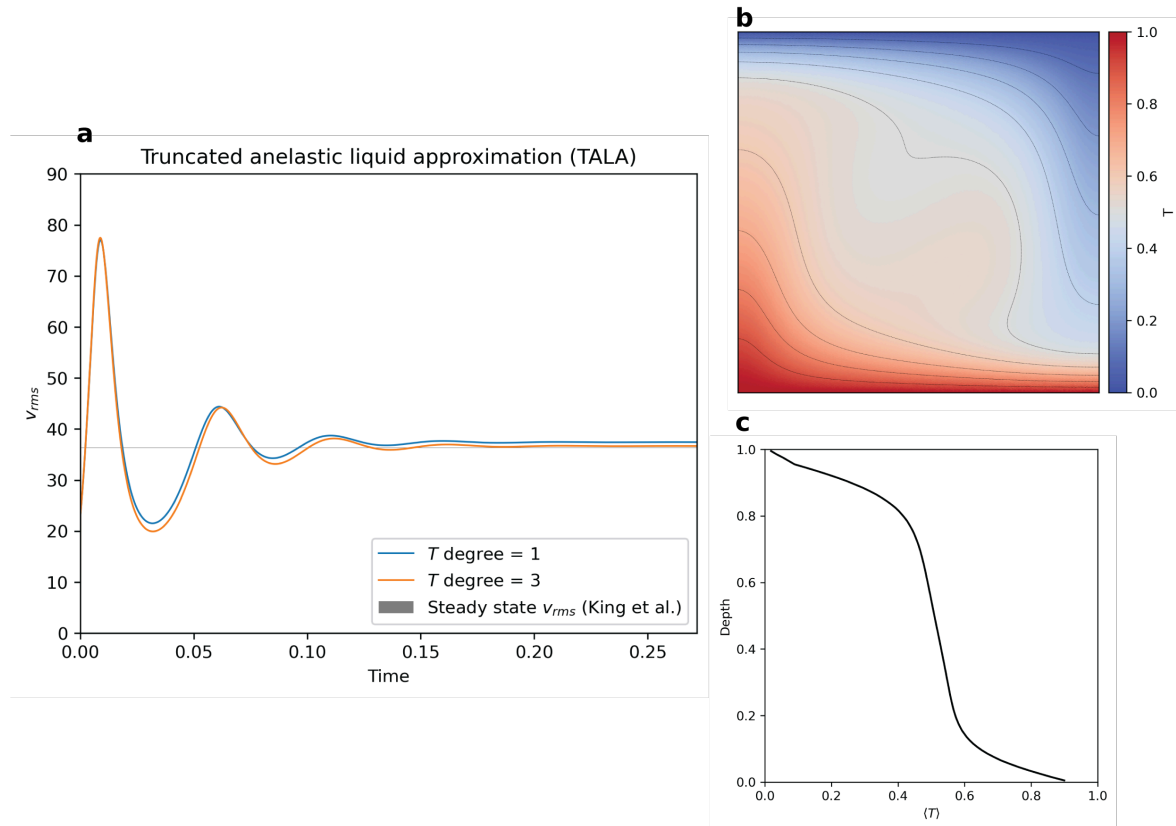


Figure 1: Figure 1. a.) Temporal evolution of the root mean squared velocity, v_{rms} , when different temperature polynomial degrees are used compared to the range of v_{rms} values listed in (King et al., 2010). b.) steady-state temperature field with temperature isolines, c.) laterally-averaged temperature vs depth.

To benchmark the implementation of TALA in Underworld3, we compare the measured steady-state root mean squared velocity, v_{rms} , to the values reported in (King et al., 2010). The compressible convection implementation in Underworld3 reaches a steady-state, as shown in the v_{rms} as a function of time plot in Figure 1a. We've also tested how close the steady-state v_{rms} is to those reported in (King et al., 2010). Using a temperature polynomial degree of 1 gets us close to the reported range with a relative difference of $\sim 3\%$. Increasing the polynomial degree to 3 further reduces the relative difference to less than 1%, as shown in Figure 1a. We also show the temperature field when steady-state is reached in Figure 1b and the laterally-averaged temperature as a function of depth in figure 1c.

The implementation codes together with the steady-state temperature, pressure, and velocity fields are found in https://github.com/underworld-community/UW3-benchmarks/blob/dev/Working/Cartesian/Ex_Convection_TALA_benchmark.py.

After this, we aim to benchmark the TALA implementation against other test cases and to implement the slightly more complicated Anelastic Liquid Approximation (ALA) formulation. We're also targeting to implement compressible convection in an annulus configuration. Stay tuned!

2. IMPORTANT IMPLEMENTATION DETAILS

If I still have your attention at this point, why not continue and check out how we did this? For the important implementation details, read ahead! =)

In compressible convection, the total temperature, T , pressure, P , and density, ρ , are expressed as a sum of a reference state (overbarred) and a perturbation (primed). That is:

$$T = \bar{T} + T'; \quad p = \bar{p} + p'; \quad \rho = \bar{\rho}(\bar{T}, \bar{p}) + \rho' \quad (1)$$

Assuming an adiabatic Adams-Williamson equation of state (Birch, 1952; King et al., 2010), and non-dimensionalising the equations, the reference states are expressed as:

$$\bar{\rho} = \rho_o \exp\left(\frac{\text{Di}}{\gamma_R}(1 - z)\right); \quad \bar{T} = T_s(\exp(\text{Di}(1 - z)) - 1) \quad (2)$$

Where ρ_o is a non-dimensionalised surface density, Di is the dissipation number, γ_R is the Grüneisen parameter, T_s is the non-dimensionalised surface temperature, and z is the non-dimensionalised depth.

3. SETTING-UP THE STOKES' EQUATIONS

Underworld, a particle-in-cell finite element modelling code, is now in its third major iteration. With the upcoming version, implementing compressible convection has never been easier. To do so, we only need to modify the Stokes' and Advection-Diffusion equations accordingly.

For compressible convection, the conservation of mass is expressed as:

$$\nabla \cdot (\bar{\rho} \vec{u}) = \nabla \cdot \vec{u} + \frac{1}{\bar{\rho}} \vec{u} \nabla \bar{\rho} = 0. \quad (3)$$

The conservation of momentum is given as

$$\nabla \cdot \left[\eta \left(\nabla \vec{u} + \nabla \vec{u}^T - \frac{2}{3} \nabla \cdot \vec{u} \mathbf{I} \right) \right] - \nabla p' - \text{Ra} \bar{\rho} \bar{g} \bar{\alpha} (T - \bar{T}) = 0, \quad (4)$$

while the deviatoric strain tensor, τ , is

$$\tau = 2\eta \dot{\epsilon} = \eta \left(\nabla \vec{u} + \nabla \vec{u}^T - \frac{2}{3} \nabla \cdot \vec{u} \mathbf{I} \right). \quad (5)$$

In these equations, $\vec{u}$, p' , T and $\dot{\epsilon}$ are the velocity, pressure, total temperature, and strain rate, respectively. In addition, $\bar{g}$, $\bar{\alpha}$, η , and Ra are the gravity, expansivity, viscosity, and Rayleigh number, respectively.

Key to the ease of implementing compressible equation is the use of Sympy in Underworld3 to symbolically define additional terms into the default systems of equations. We first define the depth-dependent reference values of density and temperature as:

```
1
2 # the non-dimensionalized reference states
3 # formulations from King et al. and Davies et al.
4 x, z = meshbox.X
5 rho_bar = rho_0*sympy.exp(Di*z/gamma_r) # rho_0 is non-dimensionalized density at surface
6 temp_bar = T_s*(sympy.exp(Di*z) - 1) # T_s is non-dimensionalized surface temperature
```

The first line stipulates that we wish to assign the symbolic representations of the meshbox coordinates to variables. The next ones indicate that we are expressing the reference functions as Sympy expressions. We can use the `display()` function to check what these symbolically look like:

```

1
2 display(rho_bar)
3 display([temp_bar])
✓ 0.0s

$$e^{0.5y}$$


$$0.091e^{0.5y} - 0.091$$


```

Next, we add the necessary terms into the Stokes equations (see Equations 3 and 4). This is done in the following lines of code, assuming that the usual methods to set-up a Stokes solver object have been done:

```

43 # adding the body force
44 # directed downwards toward z = 0
45 buoyancy_force = -Ra * rho_bar * alpha_bar * (t_soln.sym[0] - temp_bar)
46 stokes.bodyforce = sympy.Matrix([0, buoyancy_force])
47
48 # add terms into the conservation of mass according to the formulation in King et al
49 stokes.PF0 = sympy.Matrix([dot(v_soln.fn, gradient(rho_bar))/rho_bar])
50
51 stokes.stokes_problem_description()

```

The first two lines of code is for setting up the body force in both the x- and z- coordinates. The third line adds the velocity divergence term in the momentum conservation equation, while the fourth line is for adding the effect of a depth-dependent density in the conservation of mass. Note that all of these are defined symbolically. Again, using the display function to check the changes done, we get:

```

2 # verify the equations for the Stokes system
3 display(stokes._p_f0) # LHS of c.o. mass
4 display(stokes._u_f1) # LHS of c.o. momentum
5 display(stokes._u_f0) # RHS of c.o. momentum / buoyancy force

```

$$[U_{0,0}(\mathbf{x}) + 0.5U_{1,1}(\mathbf{x})]$$

$$\begin{bmatrix} -P(\mathbf{x}) + 2U_{0,0}(\mathbf{x}) & U_{0,1}(\mathbf{x}) + U_{1,0}(\mathbf{x}) \\ U_{0,1}(\mathbf{x}) + U_{1,0}(\mathbf{x}) & -P(\mathbf{x}) + 2U_{1,1}(\mathbf{x}) \end{bmatrix}$$

$$[0 \quad 10000.0 (T(\mathbf{x}) - 0.091e^{0.5y} + 0.091) e^{0.5y}]$$

Next, we set-up the Advection-Diffusion system. For the TALA formulation, the conservation of energy in terms of the total temperature, T , is given as:

$$\frac{DT}{Dt} - \frac{\text{Di}\bar{\alpha}}{\bar{c}_p} \vec{g} \cdot \vec{u}(T + T_s) = \frac{1}{\bar{\rho}\bar{c}_p} \nabla \cdot (\bar{k} \nabla T) + \frac{\text{Di}}{\text{Ra}} \frac{1}{\bar{\rho}\bar{c}_p} \phi \quad (6)$$

Where $\bar{c}_p$ is the heat capacity, $\bar{k}$ is the diffusivity, and ϕ is the viscous dissipation (Davies et al.). The viscous dissipation is calculated using the expression:

$$\phi = \frac{1}{2} \tau : \dot{\epsilon} \quad (7)$$

To add the terms associated with adiabatic heating and viscous dissipation, we do the following:

```

14 # add source terms based needed for the TALA case
15
16 #viscous dissipation term
17 comp_strainrate = stokes.strainrate -(2/3)*divergence(v_soln.fn) * sympy.eye(meshbox.dim)
18 visc_diss = 2*viscosity*comp_strainrate.norm()**2
19
20 # adiabatic heating term
21 adiab_heat = (Di*alphaBar/cp)*v_soln.sym[1]*(t_soln.sym[0] + Ts)
22
23 adv_diff.f = (Di/Ra)*(1/(rhoBar*cp))*visc_diss - adiab_heat
24
25 adv_diff.adv_diff_slcn_problem_description() # need to run this?

```

In the code snippet above, the first three lines are for symbolically defining the terms associated with the viscous dissipation and adiabatic heating (see Equations 4 and 5). The last one is for adding the two terms into the original advection-diffusion equation. Note the appropriate changes in sign as we move the terms to the right hand side of the equation and the convention we set for the direction of gravity.

The changes result to a model of isoviscous convection under the Truncated Anelastic Liquid Approximation (TALA). The rest of the code is very similar to how convection under the Boussinesq approximation is done. Have a look at the entire code and try it out yourself! Have fun learning!

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